

Exponential growth forecast of AI model size

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last updated January 15, 2026

Abstract

Based on my previous document "humanity of the third millenium under assumption of exponential growth" I extrapolate the sizes of the biggest model trained to date, and a reference size of a model that can be casually ran on consumer hardware, over the next 1000 years, once again assuming simple exponential growth.

Instead of extrapolating growth rate from current trends of AI model sizes, the growth rate is simply assumed to match that of humanity's total GDP for the biggest trained model, and that of humanity's average GDP per capita for the casual offline inference model. The starting points are set in the year 2025 with examples of Kimi K2 (1 trillion parameters) for the biggest available model and Llama3 (8 billion and 405 billion parameters) for casual and enthusiast offline inference.

The resulting model sizes are presented as charts (linear and exponential) as well as tables showing the extrapolated model size for any given year (in 5 year increments) from 2025 to 3000.

DISCLAIMER AND WARNING

This is not a scientific paper or engineering advice. This document is only intended as creative world-building and has not been peer-reviewed or approved for any practical application. Use of these ideas in real-world designs or other serious application is entirely at the reader's discretion and risk.

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1 Background and initial data

In the current AI craze, the size of AI models is growing very rapidly, and blindly extrapolating exponential growth from current data is not desired.

I am instead making an assumption that the amount of resources that can be poured into AI training is limited by the total amount of resources available, and as such follows the same growth rate as the total GDP of humanity.

In my previous document "humanity of the third millenium under assumption of exponential growth" I modeled the total global GDP of humanity as well as other data using simple exponential growth, and found that an annual growth rate of **2.88%** fits the data best.

The first AI model to be trained with **1 trillion parameters** that I am aware of was Kimi K2, released in **2025**. By necessity of how I model growth, I have to assume that after that point the growth of AI model size slowed drastically and matched the GDP growth.

The starting point is 2025, as such exponential function is set as:

$$1E + 12 \cdot 1.0288^{(x-2025)}$$

To also look at what kinds of models might be possible to run casually on consumer hardware I'll take the Llama3 model with **8 billion** parameters as a starting point in 2025. For this case, the model size should scale with GDP per capita instead of total global GDP. As with the findings of my referenced previous document, the rate of growth should in that case be 1.545% annually.

As such the second exponential function is set as:

$$8E + 9 \cdot 1.01545^{(x-2025)}$$

Lastly, NetworkChuck on youtube demonstrated enthusiast level hardware capable of running models up to 200 billion parameters in october of 2025, and earlier in february 2025 the ability to run models up to 405 billion parameters with 5 Mac Studio computers combined into a home AI inference cluster using Exo Cluster software.

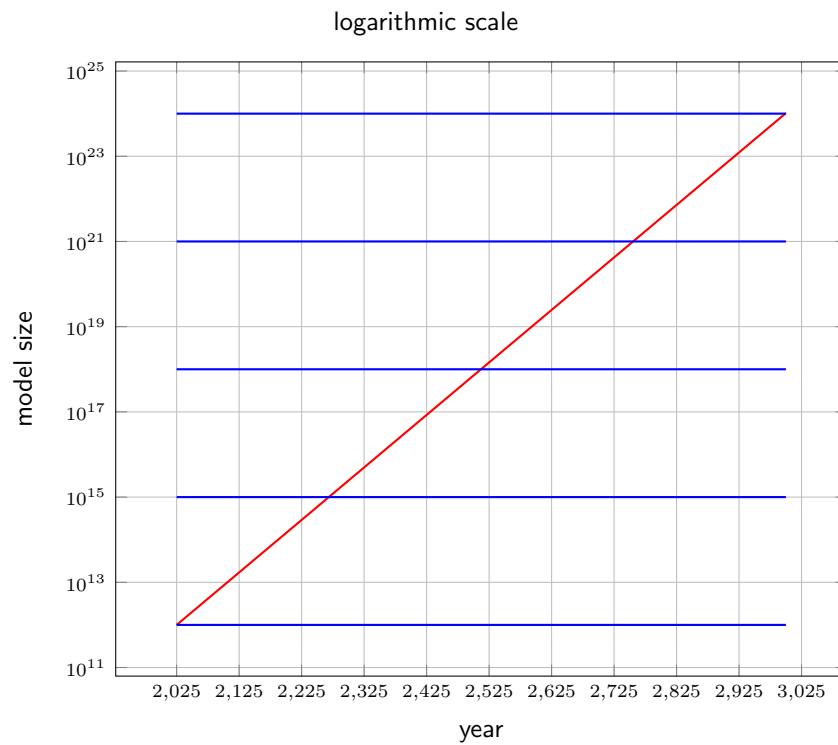
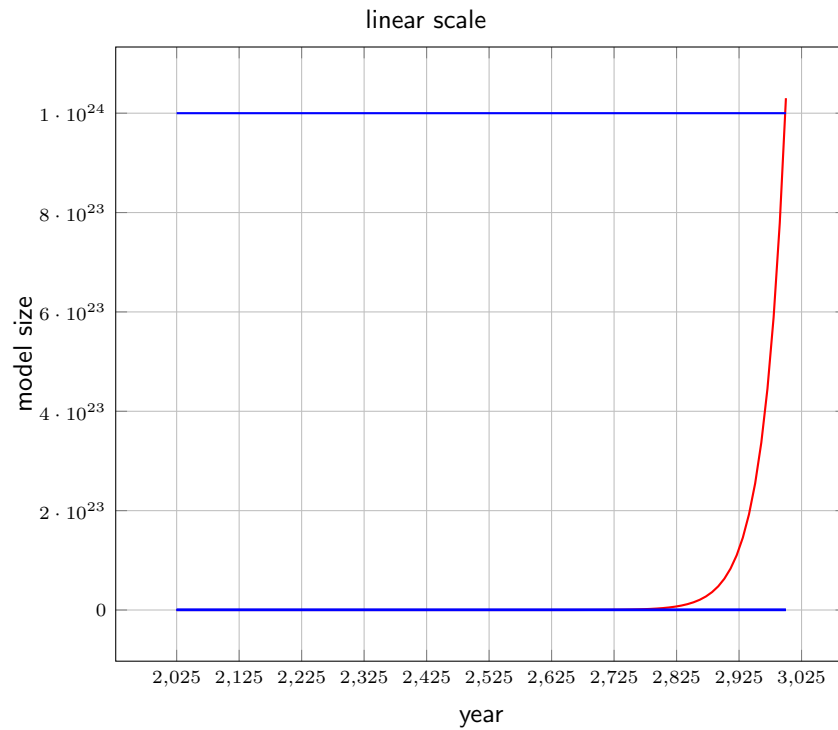
Other than the starting point, the considerations are the same, as such the third exponential function is set as:

$$405E + 9 \cdot 1.01545^{(x-2025)}$$

Keep in mind that this is not a prediction that AI model size will grow as described, but rather just a calculation of values and dates of milestones based on that assumption.

2 biggest model trained charts and data tables

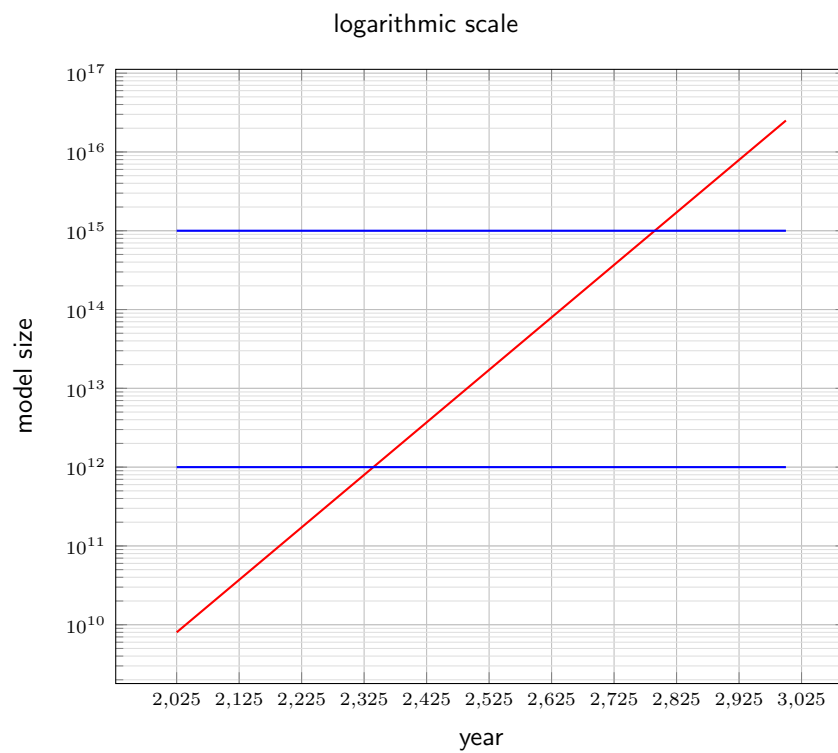
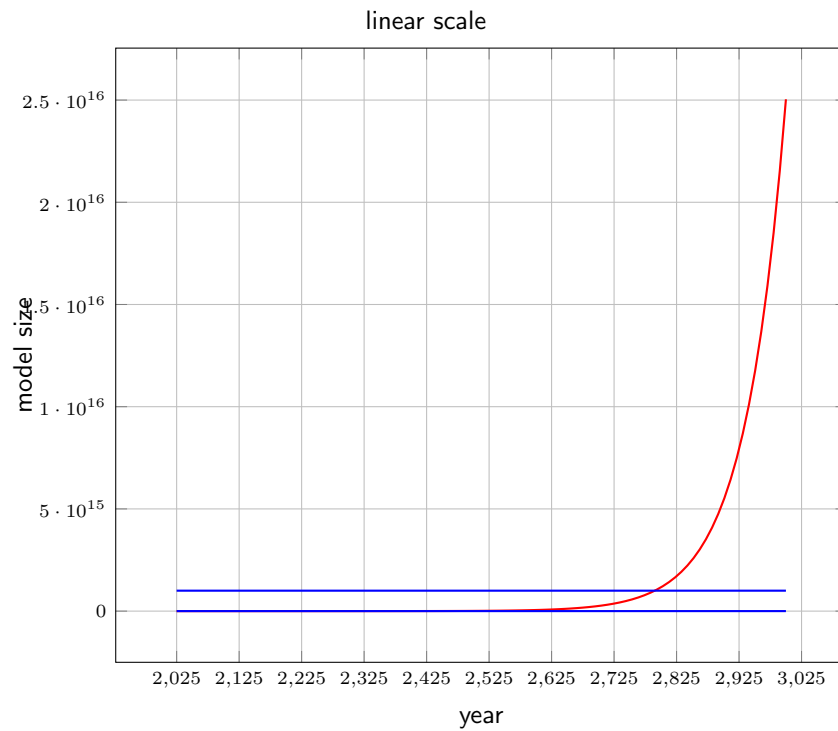
Horizontal lines and bold text mark the milestones of trillion, quadrillion, quintillion, sextillion and septillion parameters.



year	model size	year	model size	year	model size	year	model size
2025	1.0000E+12	2270	1.0497E+15	2515	1.1020E+18	2760	1.1568E+21
2030	1.1525E+12	2275	1.2099E+15	2520	1.2701E+18	2765	1.3332E+21
2035	1.3283E+12	2280	1.3944E+15	2525	1.4638E+18	2770	1.5366E+21
2040	1.5310E+12	2285	1.6071E+15	2530	1.6871E+18	2775	1.7710E+21
2045	1.7645E+12	2290	1.8523E+15	2535	1.9444E+18	2780	2.0411E+21
2050	2.0336E+12	2295	2.1348E+15	2540	2.2410E+18	2785	2.3525E+21
2055	2.3438E+12	2300	2.4604E+15	2545	2.5828E+18	2790	2.7113E+21
2060	2.7014E+12	2305	2.8357E+15	2550	2.9768E+18	2795	3.1249E+21
2065	3.1134E+12	2310	3.2683E+15	2555	3.4309E+18	2800	3.6016E+21
2070	3.5883E+12	2315	3.7668E+15	2560	3.9542E+18	2805	4.1509E+21
2075	4.1357E+12	2320	4.3414E+15	2565	4.5574E+18	2810	4.7841E+21
2080	4.7665E+12	2325	5.0036E+15	2570	5.2526E+18	2815	5.5138E+21
2085	5.4936E+12	2330	5.7669E+15	2575	6.0538E+18	2820	6.3549E+21
2090	6.3316E+12	2335	6.6465E+15	2580	6.9772E+18	2825	7.3243E+21
2095	7.2974E+12	2340	7.6604E+15	2585	8.0415E+18	2830	8.4415E+21
2100	8.4105E+12	2345	8.8289E+15	2590	9.2681E+18	2835	9.7291E+21
2105	9.6934E+12	2350	1.0176E+16	2595	1.0682E+19	2840	1.1213E+22
2110	1.1172E+13	2355	1.1728E+16	2600	1.2311E+19	2845	1.2924E+22
2115	1.2876E+13	2360	1.3517E+16	2605	1.4189E+19	2850	1.4895E+22
2120	1.4840E+13	2365	1.5578E+16	2610	1.6353E+19	2855	1.7167E+22
2125	1.7104E+13	2370	1.7955E+16	2615	1.8848E+19	2860	1.9786E+22
2130	1.9713E+13	2375	2.0694E+16	2620	2.1723E+19	2865	2.2804E+22
2135	2.2720E+13	2380	2.3850E+16	2625	2.5036E+19	2870	2.6282E+22
2140	2.6185E+13	2385	2.7488E+16	2630	2.8855E+19	2875	3.0291E+22
2145	3.0180E+13	2390	3.1681E+16	2635	3.3257E+19	2880	3.4911E+22
2150	3.4783E+13	2395	3.6514E+16	2640	3.8330E+19	2885	4.0237E+22
2155	4.0089E+13	2400	4.2083E+16	2645	4.4177E+19	2890	4.6374E+22
2160	4.6204E+13	2405	4.8502E+16	2650	5.0915E+19	2895	5.3448E+22
2165	5.3252E+13	2410	5.5901E+16	2655	5.8682E+19	2900	6.1601E+22
2170	6.1375E+13	2415	6.4428E+16	2660	6.7633E+19	2905	7.0997E+22
2175	7.0736E+13	2420	7.4255E+16	2665	7.7949E+19	2910	8.1827E+22
2180	8.1526E+13	2425	8.5582E+16	2670	8.9839E+19	2915	9.4308E+22
2185	9.3962E+13	2430	9.8636E+16	2675	1.0354E+20	2920	1.0869E+23
2190	1.0829E+14	2435	1.1368E+17	2680	1.1934E+20	2925	1.2527E+23
2195	1.2481E+14	2440	1.3102E+17	2685	1.3754E+20	2930	1.4438E+23
2200	1.4385E+14	2445	1.5101E+17	2690	1.5852E+20	2935	1.6641E+23
2205	1.6580E+14	2450	1.7404E+17	2695	1.8270E+20	2940	1.9179E+23
2210	1.9109E+14	2455	2.0059E+17	2700	2.1057E+20	2945	2.2104E+23
2215	2.2023E+14	2460	2.3119E+17	2705	2.4269E+20	2950	2.5476E+23
2220	2.5383E+14	2465	2.6645E+17	2710	2.7971E+20	2955	2.9362E+23
2225	2.9254E+14	2470	3.0710E+17	2715	3.2237E+20	2960	3.3841E+23
2230	3.3717E+14	2475	3.5394E+17	2720	3.7155E+20	2965	3.9003E+23
2235	3.8860E+14	2480	4.0793E+17	2725	4.2822E+20	2970	4.4952E+23
2240	4.4787E+14	2485	4.7015E+17	2730	4.9354E+20	2975	5.1809E+23
2245	5.1619E+14	2490	5.4187E+17	2735	5.6882E+20	2980	5.9712E+23
2250	5.9493E+14	2495	6.2452E+17	2740	6.5559E+20	2985	6.8820E+23
2255	6.8568E+14	2500	7.1979E+17	2745	7.5559E+20	2990	7.9318E+23
2260	7.9027E+14	2505	8.2958E+17	2750	8.7085E+20	2995	9.1417E+23
2265	9.1081E+14	2510	9.5612E+17	2755	1.0037E+21	3000	1.0536E+24

3 biggest model for casual offline inference

Horizontal lines and bold text mark the milestones of trillion, and quadrillion parameters.



year	model size
2025	8.0000E+09
2030	8.6374E+09
2035	9.3256E+09
2040	1.0069E+10
2045	1.0871E+10
2050	1.1737E+10
2055	1.2672E+10
2060	1.3682E+10
2065	1.4772E+10
2070	1.5949E+10
2075	1.7219E+10
2080	1.8591E+10
2085	2.0073E+10
2090	2.1672E+10
2095	2.3399E+10
2100	2.5263E+10
2105	2.7276E+10
2110	2.9449E+10
2115	3.1795E+10
2120	3.4328E+10
2125	3.7063E+10
2130	4.0016E+10
2135	4.3205E+10
2140	4.6647E+10
2145	5.0363E+10
2150	5.4376E+10
2155	5.8708E+10
2160	6.3386E+10
2165	6.8436E+10
2170	7.3889E+10
2175	7.9776E+10
2180	8.6132E+10
2185	9.2994E+10
2190	1.0040E+11
2195	1.0840E+11
2200	1.1704E+11
2205	1.2637E+11
2210	1.3643E+11
2215	1.4730E+11
2220	1.5904E+11
2225	1.7171E+11
2230	1.8539E+11
2235	2.0016E+11
2240	2.1611E+11
2245	2.3333E+11
2250	2.5192E+11
2255	2.7199E+11
2260	2.9366E+11
2265	3.1706E+11

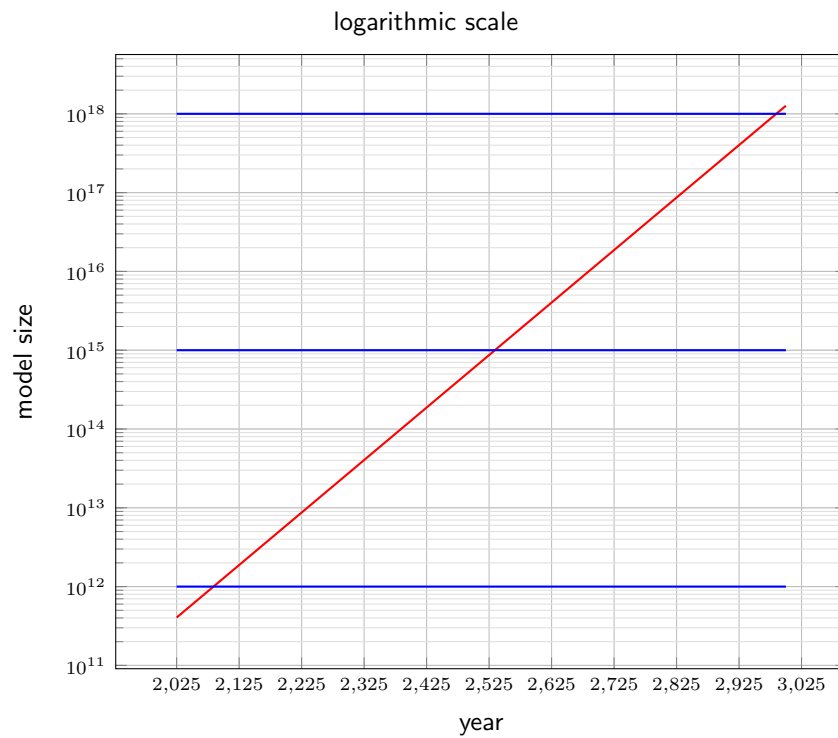
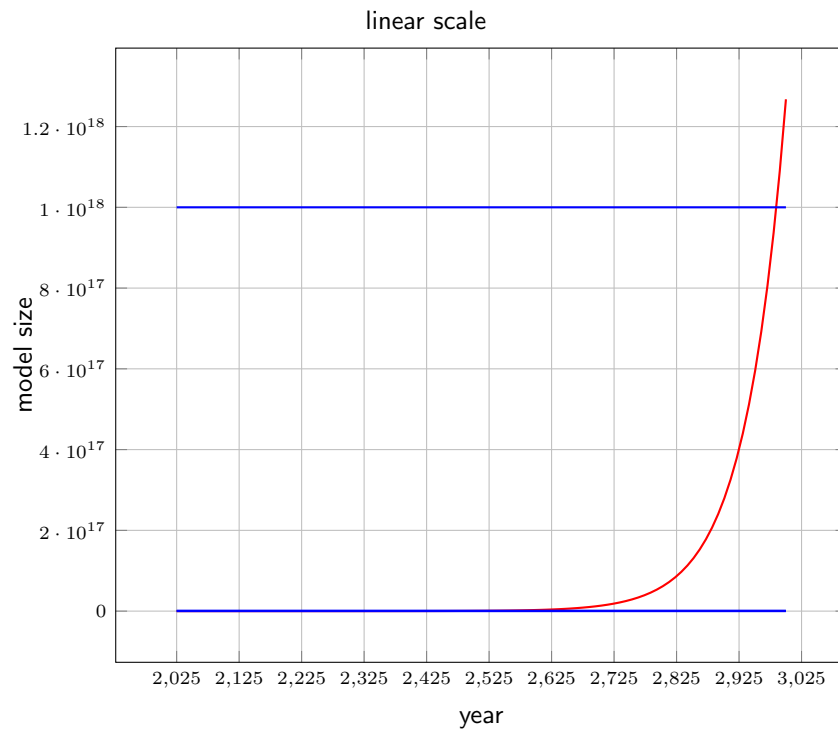
year	model size
2270	3.4232E+11
2275	3.6959E+11
2280	3.9904E+11
2285	4.3084E+11
2290	4.6516E+11
2295	5.0222E+11
2300	5.4224E+11
2305	5.8544E+11
2310	6.3208E+11
2315	6.8245E+11
2320	7.3682E+11
2325	7.9552E+11
2330	8.5891E+11
2335	9.2734E+11
2340	1.0012E+12
2345	1.0810E+12
2350	1.1671E+12
2355	1.2601E+12
2360	1.3605E+12
2365	1.4689E+12
2370	1.5859E+12
2375	1.7123E+12
2380	1.8487E+12
2385	1.9960E+12
2390	2.1551E+12
2395	2.3268E+12
2400	2.5121E+12
2405	2.7123E+12
2410	2.9284E+12
2415	3.1617E+12
2420	3.4136E+12
2425	3.6856E+12
2430	3.9792E+12
2435	4.2963E+12
2440	4.6386E+12
2445	5.0082E+12
2450	5.4072E+12
2455	5.8380E+12
2460	6.3031E+12
2465	6.8053E+12
2470	7.3475E+12
2475	7.9330E+12
2480	8.5650E+12
2485	9.2474E+12
2490	9.9842E+12
2495	1.0780E+13
2500	1.1639E+13
2505	1.2566E+13
2510	1.3567E+13

year	model size
2515	1.4648E+13
2520	1.5815E+13
2525	1.7075E+13
2530	1.8435E+13
2535	1.9904E+13
2540	2.1490E+13
2545	2.3202E+13
2550	2.5051E+13
2555	2.7047E+13
2560	2.9202E+13
2565	3.1529E+13
2570	3.4041E+13
2575	3.6753E+13
2580	3.9681E+13
2585	4.2842E+13
2590	4.6256E+13
2595	4.9941E+13
2600	5.3920E+13
2605	5.8216E+13
2610	6.2855E+13
2615	6.7863E+13
2620	7.3270E+13
2625	7.9107E+13
2630	8.5410E+13
2635	9.2215E+13
2640	9.9562E+13
2645	1.0749E+14
2650	1.1606E+14
2655	1.2531E+14
2660	1.3529E+14
2665	1.4607E+14
2670	1.5771E+14
2675	1.7027E+14
2680	1.8384E+14
2685	1.9849E+14
2690	2.1430E+14
2695	2.3137E+14
2700	2.4981E+14
2705	2.6971E+14
2710	2.9120E+14
2715	3.1440E+14
2720	3.3945E+14
2725	3.6650E+14
2730	3.9570E+14
2735	4.2722E+14
2740	4.6126E+14
2745	4.9801E+14
2750	5.3769E+14
2755	5.8053E+14

year	model size
2760	6.2679E+14
2765	6.7673E+14
2770	7.3064E+14
2775	7.8886E+14
2780	8.5171E+14
2785	9.1957E+14
2790	9.9283E+14
2795	1.0719E+15
2800	1.1573E+15
2805	1.2496E+15
2810	1.3491E+15
2815	1.4566E+15
2820	1.5727E+15
2825	1.6980E+15
2830	1.8332E+15
2835	1.9793E+15
2840	2.1370E+15
2845	2.3073E+15
2850	2.4911E+15
2855	2.6896E+15
2860	2.9038E+15
2865	3.1352E+15
2870	3.3850E+15
2875	3.6547E+15
2880	3.9459E+15
2885	4.2603E+15
2890	4.5997E+15
2895	4.9662E+15
2900	5.3619E+15
2905	5.7891E+15
2910	6.2503E+15
2915	6.7483E+15
2920	7.2860E+15
2925	7.8665E+15
2930	8.4932E+15
2935	9.1699E+15
2940	9.9005E+15
2945	1.0689E+16
2950	1.1541E+16
2955	1.2461E+16
2960	1.3453E+16
2965	1.4525E+16
2970	1.5682E+16
2975	1.6932E+16
2980	1.8281E+16
2985	1.9738E+16
2990	2.1310E+16
2995	2.3008E+16
3000	2.4841E+16

4 biggest model for enthusiast offline inference

Horizontal lines and bold text mark the milestones of trillion, quadrillion and quintillion parameters.



year	model size	year	model size	year	model size	year	model size
2025	4.0500E+11	2270	1.7330E+13	2515	7.4155E+14	2760	3.1731E+16
2030	4.3727E+11	2275	1.8711E+13	2520	8.0063E+14	2765	3.4259E+16
2035	4.7211E+11	2280	2.0202E+13	2525	8.6442E+14	2770	3.6989E+16
2040	5.0972E+11	2285	2.1811E+13	2530	9.3330E+14	2775	3.9936E+16
2045	5.5033E+11	2290	2.3549E+13	2535	1.0077E+15	2780	4.3118E+16
2050	5.9418E+11	2295	2.5425E+13	2540	1.0879E+15	2785	4.6553E+16
2055	6.4152E+11	2300	2.7451E+13	2545	1.1746E+15	2790	5.0262E+16
2060	6.9263E+11	2305	2.9638E+13	2550	1.2682E+15	2795	5.4267E+16
2065	7.4782E+11	2310	3.1999E+13	2555	1.3693E+15	2800	5.8590E+16
2070	8.0740E+11	2315	3.4549E+13	2560	1.4783E+15	2805	6.3259E+16
2075	8.7173E+11	2320	3.7301E+13	2565	1.5961E+15	2810	6.8299E+16
2080	9.4118E+11	2325	4.0273E+13	2570	1.7233E+15	2815	7.3740E+16
2085	1.0162E+12	2330	4.3482E+13	2575	1.8606E+15	2820	7.9615E+16
2090	1.0971E+12	2335	4.6947E+13	2580	2.0088E+15	2825	8.5959E+16
2095	1.1845E+12	2340	5.0687E+13	2585	2.1689E+15	2830	9.2807E+16
2100	1.2789E+12	2345	5.4725E+13	2590	2.3417E+15	2835	1.0020E+17
2105	1.3808E+12	2350	5.9086E+13	2595	2.5283E+15	2840	1.0819E+17
2110	1.4908E+12	2355	6.3793E+13	2600	2.7297E+15	2845	1.1680E+17
2115	1.6096E+12	2360	6.8876E+13	2605	2.9472E+15	2850	1.2611E+17
2120	1.7379E+12	2365	7.4364E+13	2610	3.1820E+15	2855	1.3616E+17
2125	1.8763E+12	2370	8.0288E+13	2615	3.4355E+15	2860	1.4701E+17
2130	2.0258E+12	2375	8.6685E+13	2620	3.7093E+15	2865	1.5872E+17
2135	2.1872E+12	2380	9.3592E+13	2625	4.0048E+15	2870	1.7137E+17
2140	2.3615E+12	2385	1.0105E+14	2630	4.3239E+15	2875	1.8502E+17
2145	2.5496E+12	2390	1.0910E+14	2635	4.6684E+15	2880	1.9976E+17
2150	2.7528E+12	2395	1.1779E+14	2640	5.0403E+15	2885	2.1568E+17
2155	2.9721E+12	2400	1.2718E+14	2645	5.4419E+15	2890	2.3286E+17
2160	3.2089E+12	2405	1.3731E+14	2650	5.8755E+15	2895	2.5141E+17
2165	3.4646E+12	2410	1.4825E+14	2655	6.3436E+15	2900	2.7144E+17
2170	3.7406E+12	2415	1.6006E+14	2660	6.8491E+15	2905	2.9307E+17
2175	4.0387E+12	2420	1.7281E+14	2665	7.3947E+15	2910	3.1642E+17
2180	4.3604E+12	2425	1.8658E+14	2670	7.9839E+15	2915	3.4163E+17
2185	4.7078E+12	2430	2.0145E+14	2675	8.6200E+15	2920	3.6885E+17
2190	5.0829E+12	2435	2.1750E+14	2680	9.3068E+15	2925	3.9824E+17
2195	5.4879E+12	2440	2.3483E+14	2685	1.0048E+16	2930	4.2997E+17
2200	5.9252E+12	2445	2.5354E+14	2690	1.0849E+16	2935	4.6423E+17
2205	6.3972E+12	2450	2.7374E+14	2695	1.1713E+16	2940	5.0121E+17
2210	6.9069E+12	2455	2.9555E+14	2700	1.2647E+16	2945	5.4115E+17
2215	7.4572E+12	2460	3.1910E+14	2705	1.3654E+16	2950	5.8426E+17
2220	8.0514E+12	2465	3.4452E+14	2710	1.4742E+16	2955	6.3081E+17
2225	8.6929E+12	2470	3.7197E+14	2715	1.5917E+16	2960	6.8107E+17
2230	9.3855E+12	2475	4.0161E+14	2720	1.7185E+16	2965	7.3534E+17
2235	1.0133E+13	2480	4.3360E+14	2725	1.8554E+16	2970	7.9392E+17
2240	1.0941E+13	2485	4.6815E+14	2730	2.0032E+16	2975	8.5718E+17
2245	1.1812E+13	2490	5.0545E+14	2735	2.1628E+16	2980	9.2547E+17
2250	1.2753E+13	2495	5.4572E+14	2740	2.3351E+16	2985	9.9921E+17
2255	1.3770E+13	2500	5.8920E+14	2745	2.5212E+16	2990	1.0788E+18
2260	1.4867E+13	2505	6.3614E+14	2750	2.7221E+16	2995	1.1648E+18
2265	1.6051E+13	2510	6.8683E+14	2755	2.9389E+16	3000	1.2576E+18

5 Conclusions

year	milestone
2025	first trillion-parameter AI model
2085	enthusiast hardware gets powerful enough to run trillion-parameter models
2270	first quadrillion-parameter AI model
2340	consumer hardware gets powerful enough to casually run trillion-parameter models
2515	first quintillion-parameter AI model
2535	enthusiast hardware gets powerful enough to run quadrillion-parameter models
2755	first sextillion-parameter AI model
2795	consumer hardware gets powerful enough to casually run quadrillion-parameter models
2990	enthusiast hardware gets powerful enough to run quintillion-parameter models
3000	first septillion-parameter model

The biggest model trained to date grows $\times 10$ every 81 years, and doubles every 24.4 years.

The size of models that are ran offline grows $\times 10$ every 150.2 years, and doubles every 45.2 years.